

Functional Requirements Document (FRD)

Telecom Customer Churn Analytics Dashboard & Decision Support System

Document ID: FRD-TEL-
CHURN-01

Version: 1.0 (Final)

Target Platform: Power BI /
Tableau

Domain: Enterprise Telecom BI

1. EXECUTIVE OVERVIEW & PURPOSE

This document specifies the end-to-end functional requirements for the **Telecom Customer Churn Analytics Dashboard**. The primary objective of this business intelligence solution is to empower executive leadership, churn mitigation leads, and customer success agents to monitor attrition, identify key churn drivers (contract terms, service bundles, payment channels), evaluate revenue risk, and proactively engage high-risk subscriber segments before churn occurs.

2. DASHBOARD PAGES & NAVIGATION STRUCTURE

The dashboard shall consist of four (4) distinct interactive pages, organized by decision-making scope and operational workflows:

Page 1: Executive KPI Summary

Target Audience: Chief Revenue Officer (CRO), VP of Customer Retention.

Core Purpose: Macro-level visibility into enterprise churn performance, monthly revenue loss, cohort retention trends, and overall churn health across region and customer segments.

Page 2: Drivers & Service Risk Analysis

Target Audience: Product Managers, Retention Analysts.

Core Purpose: Behavioral breakdown of churn by telecom service category (Fiber vs. DSL, Tech Support, Security add-ons), payment method friction, and contract commitment lengths.

Page 3: Predictive Risk & High-Risk Segment Modeling

Target Audience: Customer Retention Campaign Managers, ML Engineers.

Core Purpose: Predictive churn scoring, distribution of subscribers across ML risk tiers (High, Medium, Low), and proactive intervention list prioritization.

Page 4: Customer Account Drill-Through & Action Center

Target Audience: Retention Specialists, Account Executives, Call Center Leads.

Core Purpose: Micro-level inspection of individual subscriber accounts, billing history, active add-ons, churn probability score, and immediate retention offer triggers.

3. GLOBAL & PAGE-LEVEL FILTERS

All filters shall support multi-select capabilities and sync across pages unless explicitly disconnected for cohort comparison.

Filter Name	UI Control Type	Values / Selection	Scope
Contract Type	Dropdown / Multi-Select	Month-to-month, One year, Two year	Global
Internet Service Type	Horizontal Button Bar	DSL, Fiber optic, None, All	Global
Payment Method	Multi-Select Dropdown	Electronic check, Mailed check, Bank transfer (auto), Credit card (auto)	Global
Tenure Bracket	Range Slider / Category	0–12 mos, 13–24 mos, 25–48 mos, 49–72 mos	Global
Senior Citizen Status	Toggle Switch	All, Senior Citizen Only, Non-Senior Only	Page 1, 2, 3
Churn Risk Score Tier	Multi-Select Checkbox	High Risk (>70%), Medium Risk (30–70%), Low Risk (<30%)	Page 3 & 4

4. KPI DEFINITIONS & BUSINESS LOGIC

KPI Name	Display Format	Calculation Formula / Business Logic	Target / Target Variance
Overall Churn Rate	Percentage (1 Dec)	<code>COUNT(Churn = 'Yes') / COUNT(Total Customers) * 100</code>	< 18.0% (Green: <18%, Red: >22%)
Monthly Revenue Lost (MRR Churn)	Currency (\$ USD)	<code>SUM(MonthlyCharges WHERE Churn = 'Yes')</code>	Minimize MoM Growth
Total Churned Customers	Integer Count	<code>COUNTROWS(FILTER(Customers, Churn = 'Yes'))</code>	Tracking Metric
At-Risk Monthly Revenue	Currency (\$ USD)	<code>SUM(MonthlyCharges WHERE Risk_Score > 0.70 AND Churn = 'No')</code>	Proactive Mitigation KPI
Average Tenure of Churned Users	Number (Months)	<code>AVERAGE(tenure WHERE Churn = 'Yes')</code>	Target: > 18 Months
High-Risk Customer Count	Integer Count	<code>COUNT(Customers WHERE Risk_Score > 0.70 AND Churn = 'No')</code>	Target: < 500 accounts

5. VISUALIZATIONS & CHART REQUIREMENTS

Page 1 Visuals (Executive Overview)

- **KPI Summary Cards (Top Row):** Overall Churn Rate, MRR Lost, Total Churned Users, At-Risk Revenue with YoY MoM comparison flags.
- **Churn Rate by Contract Term (Clustered Bar Chart):** Compare churn percentages across Month-to-month vs 1-Year vs 2-Year contracts.
- **Tenure Cohort Survival Analysis (Line Chart):** Churn rate decay curve plotted across tenure months (0 to 72 months).
- **Monthly Revenue Loss Trend (Stacked Area Chart):** Monthly distribution of lost revenue broken down by Internet Service type over the past 12 months.

Page 2 Visuals (Drivers & Service Risk)

- **Service Adoption vs Churn Rate (100% Stacked Bar Chart):** Churn rate for users with/without Tech Support, Online Security, Device Protection, and Cloud Backup.
- **Payment Method Friction Heatmap (Matrix/Heatmap):** Payment Method vs Contract Type showing cell shade intensity based on Churn Rate (highlights Electronic Check on Month-to-Month).
- **Monthly Charges Distribution (Box Plot / Violin Plot):** Monthly bill distribution comparing Churned vs Retained customers.

Page 3 Visuals (Predictive Risk Modeling)

- **Subscriber Churn Risk Distribution (Histogram / Donut):** Customer distribution categorized into High Risk, Medium Risk, and Low Risk tiers.
- **Feature Importance Ranking (Tornado / Horizontal Bar Chart):** Top predictive drivers derived from ML models (e.g., Contract length, Tenure, Payment Method, Monthly Charges).
- **At-Risk Account Matrix (Scatter Plot):** Monthly Charges (\$) vs Tenure (Months) with dot color scaled by Churn Probability Score.

6. USER PERMISSIONS & ROLE-BASED ACCESS CONTROL (RBAC)

Data access shall be governed via Row-Level Security (RLS) based on user roles integrated with Active Directory / Azure AD groups.

ROLE NAME	TARGET PERSONA	PAGE ACCESS	DATA RESTRICTIONS (RLS)
Executive Admin	C-Suite, VP Analytics	All Pages (1, 2, 3, 4)	Full Enterprise Unrestricted Access
Regional Retention Manager	Regional Managers	All Pages (1, 2, 3, 4)	Restricted to customer accounts within assigned region/territory
Campaign Analyst	Marketing Analysts	Pages 1, 2, 3	Aggregated data only; PII (Customer Name, Phone) masked
Customer Support Rep	Call Center Agents	Page 4 Only	Can view assigned subscriber details and churn mitigation offer recommendations

7. REFRESH FREQUENCY, EXPORT, & DRILL-THROUGH REQUIREMENTS

7.1 Data Refresh Frequency

- **Batch ETL Schedule:** Daily scheduled refresh at 02:00 AM UTC for historical account balances, billing transactions, and usage logs.
- **Predictive ML Scoring Pipeline:** Weekly score update execution (Sunday 04:00 AM UTC) to re-evaluate churn risk scores across all active accounts.
- **DirectQuery Option:** Real-time event triggers for customer retention call center logging on Page 4.

7.2 Export Options & Formatting

- **Excel Data Export:** Summary tables on Page 3 & 4 must support [Export to CSV/Excel](#) (limit max 150,000 rows for security compliance).
- **PDF Report Generation:** Executive Page 1 must support formatted one-page PDF export for board presentations.
- **PowerPoint Export:** Live interactive slide embed support for quarterly business reviews.

7.3 Drill-Through Functionality

- **Global Drill-Through Path:** Right-clicking on any high-churn bar segment (e.g., Fiber Optic + Electronic Check) on Pages 1, 2, or 3 shall provide a context menu option: ["Drill-Through to Customer Action Center"](#).

- **Contextual State Preservation:** Navigating via drill-through to Page 4 shall automatically filter Page 4 to only display customer accounts matching the source visual criteria.

8. ACCEPTANCE CRITERIA

REQ ID	FEATURE CATEGORY	ACCEPTANCE CRITERIA DESCRIPTION	PRIORITY
AC-01	Performance	Dashboard pages must load fully within < 2.5 seconds upon initial opening or slice selection.	P1 - HIGH
AC-02	Data Accuracy	Total Churn Rate and Revenue Loss KPI metrics must reconcile 100% against core SAP/ERP billing database tables.	P1 - HIGH
AC-03	Security (RLS)	Support Rep users must not see unmasked PII or aggregated revenue summaries outside their assigned region.	P1 - HIGH
AC-04	Drill-Through	Drill-through from Page 3 High-Risk chart to Page 4 detail table must pass filter parameters within < 1.5 seconds.	P2 - MED
AC-05	Mobile View	Page 1 Executive KPI section must render cleanly on tablet/mobile screens without horizontal scrolling.	P3 - LOW